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VEHICLE FRAME ASSEMBLY, ENERGY STORAGE STRUCTURE, AND VEHICLE

Abstract

A vehicle includes a vehicle frame assembly and an energy storage structure. The vehicle frame assembly includes a vehicle frame front section, a battery pack mounting frame and a vehicle frame rear section which are sequentially connected in the length direction of a vehicle body. The battery pack mounting frame is used for mounting a battery pack.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present disclosure is a bypass continuation of PCT International Application No. PCT/CN2023/116048, filed on Aug. 31, 2023, which claims priority to Chinese Patent Application No. 202310363241.1 filed on Mar. 29, 2023 and the invention title of “VEHICLE FRAME ASSEMBLY, ENERGY STORAGE STRUCTURE, AND VEHICLE”, which are incorporated herein by reference in their entireties.

FIELD

[0002] The present disclosure relates to a vehicle frame assembly, an energy storage structure and a vehicle.

BACKGROUND

[0003] In the related art, a battery bracket is usually adopted to suspend and install the battery pack on the front and rear cross-beams of the vehicle frame. At this time, since the vehicle frame limits the installation space of the battery pack, and the battery pack, the battery bracket, the longitudinal beams on both sides, and the middle cross-beam are all independent components, the installation is cumbersome, the installation accuracy requirements are high, the weight is large, and the cost is high.

SUMMARY

[0004] The technical problem to be solved by the present disclosure is as follows: In the existing power battery installation structure, the battery pack, the battery bracket, the longitudinal beams on both sides, and the middle cross-beam are all independent components, which leads to a large weight of the vehicle frame. To solve this issue, a vehicle frame assembly, an energy storage structure, and a vehicle are provided.

[0005] To solve the foregoing technical problem, on the one hand, an embodiment of the present disclosure provides a vehicle frame assembly, which includes a vehicle frame front section, a battery pack mounting frame, and a vehicle frame rear section that are sequentially connected along the length direction of the vehicle body. The battery pack mounting frame is used to install the battery pack.

[0006] In an embodiment, along the width direction of the vehicle body, the size of the battery pack mounting frame is not less than the size of the vehicle frame front section or the vehicle frame rear section. In other words, along the width direction of the vehicle body, the width of the battery pack mounting frame is not less than the width of the vehicle frame front section or the vehicle frame rear section. In an embodiment, the vehicle frame assembly further includes a connection component. The connection component connects the battery pack mounting frame with the vehicle frame front section and/or connects the battery pack mounting frame with the vehicle frame rear section.

[0007] In an embodiment, the vehicle frame front section includes a front cross-beam and two front longitudinal beams. The front cross-beam is connected between the two front longitudinal beams; the vehicle frame rear section includes a rear cross-beam and two rear longitudinal beams. The rear cross-beam is connected between the two rear longitudinal beams; the connection component includes a first connector and a second connector. The first connector connects the battery pack mounting frame with the front cross-beam and/or connects the battery pack mounting frame with

the rear cross-beam, and the second connector connects the battery pack mounting frame with the front longitudinal beam and/or connects the battery pack mounting frame with the rear longitudinal beam.

[0008] In an embodiment, the vehicle frame assembly further includes a cover plate. The cover plate covers the top of the battery pack mounting frame, and the second connector is arranged with an upward-turned edge for connecting with the cover plate.

[0009] In an embodiment, the connection component further includes a third connector. The third connector is arranged on the outer side of the second connector and connects the second connector and the battery pack mounting frame.

[0010] The third connector is arranged with a side-turned edge. The side-turned edge is arranged outward along the width direction of the vehicle body to strengthen the strength of the third connector along the width direction of the vehicle body.

[0011] In an embodiment, the battery pack mounting frame is formed by a plurality of longitudinal beams, a plurality of cross-beams, and a plurality of vertical beams intersecting with each other to form a space for accommodating at least one layer of battery packs.

[0012] In an embodiment, the battery pack mounting frame further includes a plurality of diagonal beams and a plurality of mounting frame reinforcing plates. The diagonal beams are arranged between the intersection points of two longitudinal beams and cross-beams; the mounting frame reinforcing plates are arranged at the bottom and/or the side of the battery pack mounting frame.

[0013] According to the vehicle frame assembly provided by the embodiment of the present disclosure, it includes a vehicle frame front section, a battery pack mounting frame, and a vehicle frame rear section that are sequentially connected along the length direction of the vehicle body. The battery pack mounting frame is used to install the battery pack. In this way, the present disclosure is equivalent to dividing the traditional integral vehicle frame into three sections, and the vehicle frame middle section is combined with the battery pack mounting frame. Compared with the prior art, the battery pack mounting frame of the vehicle frame assembly of the present disclosure constitutes the vehicle frame middle section, which can reduce the number of components and the weight of the vehicle frame, is beneficial to the lightweight of the whole vehicle, and improves the vehicle's endurance. In addition, the battery pack is installed on the battery pack mounting frame that constitutes the vehicle frame middle section. When a collision occurs, the battery pack participates in the force-bearing of the whole vehicle, which can improve the torsional stiffness of the whole vehicle.

[0014] On the other hand, an embodiment of the present disclosure provides an energy storage structure, which includes a battery pack and the foregoing vehicle frame assembly. The battery pack is installed on the battery pack mounting frame.

[0015] In an embodiment, at least one of the two sides of the battery pack along the vehicle body direction is arranged with a notch. The notch is used for the pipeline layout on the vehicle.

[0016] Furthermore, an embodiment of the present disclosure provides a vehicle, which includes the foregoing energy storage structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a schematic diagram of an energy storage structure provided by an embodiment of the present disclosure;

[0018] FIG. 2 is an exploded view of a vehicle frame assembly provided by an embodiment of the present disclosure;

[0019] FIG. 3 is a schematic diagram of the vehicle frame assembly after removing the cover plate provided by an embodiment of the present disclosure;

[0020] FIG. 4 is a partially enlarged view of the connection part between the vehicle frame front section and the vehicle frame middle section of the vehicle frame assembly provided by an embodiment of the present disclosure;

[0021] FIG. 5 is a partially enlarged view of the connection part between the front end of the left installation reinforcing plate of the battery pack bracket of the vehicle frame assembly and the battery pack bracket provided by an embodiment of the present disclosure;

[0022] FIG. 6 is a schematic diagram of the energy storage structure after removing the cover plate provided by an embodiment of the present disclosure;

[0023] FIG. 7 is an enlarged view of the part a in FIG. 6;

[0024] FIG. 8 is an enlarged view of the part b in FIG. 3;

[0025] FIG. 9 is a schematic diagram of the battery pack of the energy storage structure provided by an embodiment of the present disclosure;

[0026] FIG. 10 is a schematic diagram of an energy storage structure provided by another embodiment of the present disclosure;

[0027] The reference numerals in the specification are as follows: [0028] 1. battery pack; 11. notch;

[0029] 2. vehicle frame front section; 21. front cross-beam; 211. front cross-beam main body; 212.

front connector; 22. front longitudinal beam; [0030] 3. vehicle frame rear section; 31. rear cross-

beam; 311. rear cross-beam main body; 312. rear connector; 32. rear longitudinal beam; [0031] 4.

battery pack mounting frame; 411. front upper cross-beam; 412. front lower cross-beam; 413. left

front vertical beam; 414. right front vertical beam; 415. rear upper cross-beam; 416. rear lower

cross-beam; 417. left rear vertical beam; 418. right rear vertical beam; 419. left upper longitudinal

beam; 4110. left lower longitudinal beam; 4111. right upper longitudinal beam; 4112. right lower

longitudinal beam; 4113. top middle longitudinal beam; 4114. bottom middle longitudinal beam;

4115. middle partition frame; 41151. top middle cross-beam; 41152. left middle vertical beam;

41153. right middle vertical beam; 41154. bottom supporting cross-beam; 411541. first supporting

cross-beam; 411542. second supporting cross-beam; 41155. connecting vertical beam; 41156. left

diagonal beam; 41157. right diagonal beam; 4116. front supporting cross-beam; 4117. rear

supporting cross-beam; 4118. front middle cross-beam; 4119. rear middle cross-beam; 42. left

installation reinforcing plate of the battery pack bracket; 43. right installation reinforcing plate of

the battery pack bracket; 441. left front connector; 442. right front connector; 443. left rear

connector; 444. right rear connector; [0032] 5. connection component; 51. first connector; 511.

avoidance hole; 52. second connector; 521. upward-turned edge; 522. left front inner reinforcing

plate; 523. right front inner reinforcing plate; 524. left rear inner reinforcing plate; 525. right rear

inner reinforcing plate; 526. downward-turned edge; 53. third connector; 531. side-turned edge;

532. left front outer reinforcing plate; 533. right front outer reinforcing plate; 534. left rear outer

reinforcing plate; 535. right rear outer reinforcing plate; [0033] 6. cover plate.

DETAILED DESCRIPTION

[0034] In order to make the technical problems, technical solutions, and beneficial effects solved

by the present disclosure clearer, the following further describes the present disclosure in detail

with reference to the accompanying drawings and embodiments. It should be understood that the

specific embodiments described herein are only used to explain the present disclosure, and are not

intended to limit the present disclosure.

[0035] The front, rear, up, down, left, and right in the present disclosure are the vehicle orientations

understood in a conventional sense, as shown in the coordinates in FIG. 1.

[0036] Referring to FIG. 1 to FIG. 9, an embodiment of the present disclosure provides an energy

storage structure, the energy storage structure includes a battery pack 1 and a vehicle frame

assembly. The vehicle frame assembly includes a vehicle frame front section 2, a battery pack

mounting frame 4, and a vehicle frame rear section 3 that are sequentially connected along the

length direction of the vehicle body. The battery pack mounting frame 4 is used to install the

battery pack. In other words, the battery pack 1 is integrated on the battery pack mounting frame 4

that constitutes the vehicle frame middle section.

[0037] According to the vehicle frame assembly and energy storage structure provided by the embodiment of the present disclosure, it is equivalent to dividing the traditional integral vehicle frame into three sections, and the vehicle frame middle section is combined with the battery pack mounting frame **4**. Compared with the prior art, the battery pack mounting frame **4** of the vehicle frame assembly of the present disclosure constitutes the vehicle frame middle section, reducing the girder and cross-beam of the vehicle frame middle section and the connection components used for connecting the girder and cross-beam. Compared with the prior art, the number of parts of the vehicle frame assembly can be reduced by more than 10%, and the weight is reduced by more than 10%, which is beneficial to the lightweight of the whole vehicle and improves the vehicle's endurance. In addition, the battery pack is installed on the battery pack mounting frame **4** that constitutes the vehicle frame middle section. When a collision occurs, the battery pack participates in the force-bearing of the whole vehicle, which can improve the torsional stiffness of the whole vehicle. For example, in the prior art, the torsional stiffness of the whole vehicle is only 7 to 9 modes, while after applying the vehicle frame assembly of the embodiment of the present disclosure, the torsional stiffness of the whole vehicle can reach 14 to 20 modes.

[0038] The vehicle frame assembly further includes a connection component **5**. The connection component **5** connects the battery pack mounting frame **4** with the vehicle frame front section **2** and/or connects the battery pack mounting frame **4** with the vehicle frame rear section **3**.

[0039] In one embodiment, along the width direction of the vehicle body, the size of the battery pack mounting frame **4** is not less than the size of the vehicle frame front section **2** or the vehicle frame rear section **3**. In other words, along the width direction of the vehicle body, the width of the battery pack mounting frame **4** is not less than the width of the vehicle frame front section or the vehicle frame rear section. In this way, the battery pack mounting frame **4** has sufficient accommodating space to install more batteries and improve the vehicle's endurance.

[0040] In one embodiment, referring to FIG. **1**, FIG. **6** to FIG. **8**, the vehicle frame front section **2** includes a front cross-beam **21** and two front longitudinal beams **22**. The front cross-beam **21** is connected between the two front longitudinal beams **22**; the vehicle frame rear section **3** includes a rear cross-beam **31** and two rear longitudinal beams **32**. The rear cross-beam **31** is connected between the two rear longitudinal beams **32**. The connection component **5** includes a first connector **51** and a second connector **52**. The first connector **51** connects the battery pack mounting frame **4** with the front cross-beam **21** and/or connects the battery pack mounting frame **4** with the rear cross-beam **31**, and the second connector **52** connects the battery pack mounting frame **4** with the front longitudinal beam **22** and/or connects the battery pack mounting frame **4** with the rear longitudinal beam **32**.

[0041] Specifically, referring to FIG. **8** and FIG. **9**, the front cross-beam **21** includes a front cross-beam main body **211** and a front connector **212**. The front connector **212** is fixed on the inner side of the front longitudinal beam **22**. The left end of the front cross-beam main body **211** is fixedly connected to the left front connector **212**, and the right end of the front cross-beam main body **211** is fixedly connected to the right front connector **212**. The rear cross-beam **31** includes a rear cross-beam main body **311** and a rear connector **312**. The rear connector **312** is fixed on the inner side of the rear longitudinal beam **32**. The left end of the rear cross-beam main body **311** is fixedly connected to the left rear connector **312**, and the right end of the rear cross-beam main body **311** is fixedly connected to the right rear connector **312**.

[0042] The rear longitudinal beam **32** is a U-shaped beam structure with an opening inward, the rear cross-beam main body **311** is a U-shaped beam structure with an opening facing backward, the rear connector **312** is a U-shaped beam structure with an opening inward. The rear connector **312** is embedded in the rear longitudinal beam **32**, and the rear cross-beam main body **311** is inserted into the rear connector **312**. The first connector **51** is in the shape of a plate, and the first connector **51** on the rear side is clamped between the front side of the rear cross-beam main body **311** and the

rear side of the battery pack mounting frame **4**.

[0043] Similarly, the front longitudinal beam **22** is a U-shaped beam structure with an opening inward, the front cross-beam main body **211** is a U-shaped beam structure with an opening facing backward, the front connector **212** is a U-shaped beam structure with an opening inward. The front connector **212** is embedded in the front longitudinal beam **22**, and the front cross-beam main body **211** is inserted into the front connector **212**. The first connector **51** on the rear side is clamped between the rear side of the front cross-beam main body **211** and the front side of the battery pack mounting frame **4**.

[0044] In some embodiments, the battery pack mounting frame **4** is connected to the vehicle frame front section **2** through the connection component **5**, and the battery pack mounting frame **4** is further connected to the vehicle frame rear section **3** through the connection component **5**. In other words, two connection components **5** are provided. The first connector **51** of one connection component **5** connects the battery pack mounting frame **4** with the front cross-beam **21**, and the first connector **51** of the other connection component **5** connects the battery pack mounting frame **4** with the rear cross-beam **31**. The second connector **52** of one connection component **5** connects the battery pack mounting frame **4** with the front longitudinal beam **22**, and the second connector **52** of the other connection component **5** connects the battery pack mounting frame **4** with the rear longitudinal beam **32**.

[0045] In one embodiment, referring to FIG. **1** and FIG. **2**, the vehicle frame assembly further includes a cover plate **6**. The second connector **52** is arranged with an upward-turned edge **521** for connecting with the cover plate **6**, and the upward-turned edge **521** extends outward along the left-right direction of the vehicle body. The cover plate **6** covers the top of the battery pack mounting frame **4** and is used for the top protection of the battery pack **1**. In some embodiments, the cover plate **6** is fixed on the upward-turned edge **521** of the second connector **52** by fasteners (such as bolts).

[0046] When the cover plate **6** is connected to the upward-turned edge **521** of the second connector **52** of the battery pack mounting frame **4** by bolts, if the bolts are too short, their bending resistance and stiffness are poor, and if the bolts are too long, they will occupy a large vertical space.

Therefore, in some embodiments, the fixed length of the bolts is 300 to 500 mm.

[0047] Of course, in other embodiments, the cover plate **6** can further be riveted or welded to the top of the battery pack mounting frame **4**.

[0048] In one embodiment, referring to FIG. **4**, FIG. **7** and FIG. **8**, the connection component **5** further includes a third connector **53**. The third connector **53** is arranged on the outer side of the second connector **52** and connects the second connector **52** and the battery pack mounting frame **4**. The third connector **53** is arranged with a side-turned edge **531**, and the side-turned edge **531** is arranged outward along the width direction of the vehicle body (the left-right direction in FIG. **1**) to strengthen the strength of the third connector **531** along the width direction of the vehicle body.

[0049] In one embodiment, referring to FIG. **3** and FIG. **4**, two second connectors **52** are arranged on the front side and two on the rear side. The second connectors **52** on the front side are a left front inner reinforcing plate **522** and a right front inner reinforcing plate **523**, and the second connectors **52** on the rear side are a left rear inner reinforcing plate **524** and a right rear inner reinforcing plate **525**; the left front inner reinforcing plate **522** is closely fixed on the left side of the vehicle frame front section **2** (i.e., the left side of the left front longitudinal beam **21**), and the right front inner reinforcing plate **523** is closely fixed on the right side of the vehicle frame front section **2** (i.e., the right side of the right front longitudinal beam **21**); the left rear inner reinforcing plate **524** is closely fixed on the left side of the vehicle frame rear section **3** (i.e., the left side of the left rear longitudinal beam **31**), and the right rear inner reinforcing plate **525** is closely fixed on the right side of the vehicle frame rear section **3** (i.e., the right side of the right rear longitudinal beam **31**). In this way, the connection strength between the front side of the battery pack mounting frame **4** and the vehicle frame front section **2** can be further increased through the left front inner

reinforcing plate 522 and the right front inner reinforcing plate 523, and the connection strength between the rear side of the battery pack mounting frame 4 and the vehicle frame rear section 3 can be further increased through the left rear inner reinforcing plate 524 and the right rear inner reinforcing plate 525.

[0050] In one embodiment, referring to FIG. 3 and FIG. 4, two third connectors 53 are arranged on the front side and two on the rear side. The third connectors 53 on the front side are a left front outer reinforcing plate 532 and a right front outer reinforcing plate 533, and the third connectors 53 on the rear side are a left rear outer reinforcing plate 534 and a right rear outer reinforcing plate 535; the left front outer reinforcing plate 532 is closely fixed on the left side of the left front inner reinforcing plate 522, and the left front outer reinforcing plate 532 is fixedly connected between the left side of the vehicle frame front section 2 (i.e., the left front longitudinal beam 21) and the front end of the battery pack mounting frame 4 by bolts, riveting or welding; the right front outer reinforcing plate 533 is closely fixed on the right side of the right front inner reinforcing plate 523, and the right front outer reinforcing plate 533 is fixedly connected between the right side of the vehicle frame front section 2 (i.e., the right front longitudinal beam 21) and the front end of the battery pack mounting frame 4 by bolts, riveting or welding; the left rear outer reinforcing plate 534 is closely fixed on the left side of the left rear inner reinforcing plate 524, and the left rear outer reinforcing plate 534 is fixedly connected between the left side of the vehicle frame rear section 3 (the left rear longitudinal beam 31) and the rear end of the battery pack mounting frame 4 by bolts, riveting or welding; the right rear outer reinforcing plate 535 is closely fixed on the right side of the right rear inner reinforcing plate 525, and the right rear outer reinforcing plate 535 is fixedly connected between the right side of the vehicle frame rear section 3 (the right rear longitudinal beam 31) and the rear end of the battery pack mounting frame 4 by bolts, riveting or welding. In this way, the connection strength between the front side of the battery pack mounting frame 4 and the vehicle frame front section 2 can be increased through the left front outer reinforcing plate 532 and the right front outer reinforcing plate 533, and the connection strength between the rear side of the battery pack mounting frame 4 and the vehicle frame rear section 3 can be increased through the left rear outer reinforcing plate 534 and the right rear outer reinforcing plate 535.

[0051] In one embodiment, referring to FIG. 3 and FIG. 6, the battery pack mounting frame 4 is a single-layer frame structure formed by a plurality of longitudinal beams, a plurality of cross-beams and a plurality of vertical beams intersecting with each other to form a space for accommodating one layer of battery packs 1. The battery pack mounting frame 4 is approximately square, and there are multiple battery packs 1, and the multiple battery packs 1 are arranged in a single layer in the up-down direction. For example, in the embodiment shown in FIG. 1, there are two battery packs 1, and the two battery packs 1 are arranged in the front-rear direction (front and rear arrangement).

[0052] The longitudinal beams extend in the front-rear direction of the vehicle body, the cross-beams extend in the left-right direction of the vehicle body, and the vertical beams extend in the up-down direction of the vehicle body.

[0053] However, in other embodiments, corresponding to the single-layer frame structure of the battery pack mounting frame 4, only one battery pack 1 can further be provided.

[0054] Of course, in other embodiments, corresponding to the single-layer frame structure of the battery pack mounting frame 4, multiple battery packs 1 can further be provided. The multiple battery packs 1 are arranged left and right, or the multiple battery packs 1 are arranged in multiple rows and columns (rectangular array arrangement).

[0055] The single-layer frame structure of the battery pack mounting frame 4 can be used for vehicle models with a small number of required battery packs 1. Or for vehicle models with a small installation space for the battery packs 1 in the vertical direction.

[0056] Of course, in other embodiments, when there are multiple battery packs 1, the battery pack mounting frame 4 can further be a multi-layer frame structure formed by a plurality of longitudinal

beams and a plurality of cross-beams intersecting with each other to form a space for accommodating multiple layers of battery packs **1**. The multiple battery packs **1** are arranged in multiple layers in the up-down direction. At this time, each layer of the multi-layer frame structure is used to install the battery packs **1** of the corresponding layer. One or more battery packs **1** are arranged in each layer. When there are multiple battery packs **1** in each layer, the multiple battery packs **1** in each layer are arranged front and rear or left and right, or the multiple battery packs **1** in each layer are arranged in multiple rows and columns, or the multiple battery packs **1** in some layers are arranged front and rear or left and right, and the multiple battery packs **1** in some layers are arranged in multiple rows and columns.

[0057] The multi-layer frame structure of the battery pack mounting frame **4** can be used for vehicle models with a large number of required battery packs **1**. Or for vehicle models with a large installation height of the battery packs **1** in the vertical direction.

[0058] In summary, the energy storage structure of the embodiment of the present disclosure can arrange multiple battery packs in the X direction (the front-rear direction of the vehicle body), the Y direction (the left-right direction of the vehicle body), and the Z direction (the up-down direction of the vehicle body) according to the size of the whole vehicle.

[0059] In one embodiment, referring to FIG. 2 and FIG. 3, the battery pack mounting frame **4** is a square frame structure. The multiple cross-beams of the battery pack mounting frame **4** include a front upper cross-beam **411**, a front lower cross-beam **412**, a rear upper cross-beam **415** and a rear lower cross-beam **416**, the multiple longitudinal beams of the battery pack mounting frame **4** include a left front vertical beam **413**, a right front vertical beam **414**, a right upper longitudinal beam **4111** and a right lower longitudinal beam **4112**, the multiple vertical beams of the battery pack mounting frame **4** include a left front vertical beam **413**, a right front vertical beam **414**, a left rear vertical beam **417** and a right rear vertical beam **418**. The front upper cross-beam **411**, the front lower cross-beam **412**, the rear upper cross-beam **415** and the rear lower cross-beam **416** extend in the left-right direction of the vehicle body, the left upper longitudinal beam **419**, the left lower longitudinal beam **4110**, the right upper longitudinal beam **4111** and the right lower longitudinal beam **4112** extend in the front-rear direction of the vehicle body, and the left front vertical beam **413**, the right front vertical beam **414**, the left rear vertical beam **417** and the right rear vertical beam **418** extend in the up-down direction of the vehicle body; the front upper cross-beam **411** is fixedly connected between the front end of the left upper longitudinal beam **419** and the front end of the right upper longitudinal beam **4111**, the front lower cross-beam **412** is fixedly connected between the front end of the left lower longitudinal beam **4110** and the front end of the right lower longitudinal beam **4112**, the left front vertical beam **413** is fixedly connected between the front end of the left upper longitudinal beam **419** and the front end of the left lower longitudinal beam **4110**, the right front vertical beam **414** is fixedly connected between the front end of the right upper longitudinal beam **4111** and the front end of the right lower longitudinal beam **4112**, the rear upper cross-beam **415** is fixedly connected between the rear end of the left upper longitudinal beam **419** and the rear end of the right upper longitudinal beam **4111**, the rear lower cross-beam **416** is fixedly connected between the rear end of the left lower longitudinal beam **4110** and the rear end of the right lower longitudinal beam **4112**, the left rear vertical beam **417** is fixedly connected between the rear end of the left upper longitudinal beam **419** and the rear end of the left lower longitudinal beam **4110**, and the right rear vertical beam **418** is fixedly connected between the rear end of the right upper longitudinal beam **4111** and the rear end of the right lower longitudinal beam **4112**; the front upper cross-beam **411**, the left front vertical beam **413**, the front lower cross-beam **412** and the right front vertical beam **414** are connected in sequence to form a front closed-loop frame structure to enhance the front strength of the battery pack mounting frame **4**. The rear upper cross-beam **415**, the left rear vertical beam **417**, the rear lower cross-beam **416** and the right rear vertical beam **418** are connected in sequence to form a rear closed-loop frame structure to enhance the rear strength of the battery pack mounting frame **4**. The front upper cross-beam **411**, the left upper

longitudinal beam **419**, the rear upper cross-beam **415** and the right upper longitudinal beam **4111** are connected in sequence to form a top closed-loop frame structure to enhance the top strength of the battery pack mounting frame **4**. The front lower cross-beam **412**, the left lower longitudinal beam **4110**, the rear lower cross-beam **416** and the right lower longitudinal beam **4112** are connected in sequence to form a bottom closed-loop frame structure to enhance the bottom strength of the battery pack mounting frame **4**. The left upper longitudinal beam **419**, the left front vertical beam **413**, the left lower longitudinal beam **4110** and the left rear vertical beam **418** are connected in sequence to form a left closed-loop frame structure to enhance the left strength of the battery pack mounting frame **4**. The right upper longitudinal beam **4111**, the right front vertical beam **414**, the right lower longitudinal beam **4112** and the right rear vertical beam **418** are connected in sequence to form a right closed-loop frame structure to enhance the right strength of the battery pack mounting frame **4**.

[0060] In this way, the front closed-loop frame structure, the rear closed-loop frame structure, the top closed-loop frame structure, the bottom closed-loop frame structure, the left closed-loop frame structure and the right closed-loop frame structure can enhance the strength of the battery pack mounting frame **4** in all directions and enhance the overall strength of the battery pack mounting frame **4**, so that the battery pack **1** can be well protected from all directions when the vehicle collides.

[0061] In one embodiment, referring to FIG. 2 and FIG. 3, the multiple longitudinal beams of the battery pack mounting frame **4** further include a top middle longitudinal beam **4113** and a bottom middle longitudinal beam **4114**. The top middle longitudinal beam **4113** and the bottom middle longitudinal beam **4114** extend in the front-rear direction of the vehicle body; there is one top middle longitudinal beam **4113** or multiple top middle longitudinal beams **4113** are arranged in parallel, and there is one bottom middle longitudinal beam **4114** or multiple bottom middle longitudinal beams **4114** are arranged in parallel. For example, in the embodiment shown in FIG. 3, there are two top middle longitudinal beams **4113** and two bottom middle longitudinal beams **4114**, and the top middle longitudinal beam **4113** is aligned with the corresponding bottom middle longitudinal beam **4114** in the vertical direction. The top middle longitudinal beam **4113** is connected between the front upper cross-beam **411** and the rear upper cross-beam **415**, and the top middle longitudinal beam **4113** is located between the left upper longitudinal beam **419** and the right upper longitudinal beam **4111** in the left-right direction of the vehicle frame assembly; the bottom middle longitudinal beam **4114** is connected between the front lower cross-beam **412** and the rear lower cross-beam **416**, and the bottom middle longitudinal beam **4114** is located between the left lower longitudinal beam **4110** and the right lower longitudinal beam **4112** in the left-right direction of the vehicle frame assembly. The top middle longitudinal beam **4113** can enhance the top strength of the battery pack mounting frame **4**, and the bottom middle longitudinal beam **4114** can enhance the bottom strength of the battery pack mounting frame **4**. Furthermore, the overall strength of the battery pack mounting frame **4** is further enhanced.

[0062] In one embodiment, referring to FIG. 2 and FIG. 3, the battery pack mounting frame **4** further includes a middle partition frame **4115** for separating the battery packs **1** in the front-rear direction. One or more middle partition frames **4115** are provided to divide the interior of the battery pack mounting frame **4** into multiple battery pack installation spaces arranged in the front-rear direction. In this way, each battery pack installation space can be used to place a battery pack **1**, which can increase the number of battery packs **1**.

[0063] For example, in the embodiment shown in FIG. 2, there is one middle partition frame **4115** that divides the interior of the battery pack mounting frame **4** into two battery pack installation spaces arranged in the front-rear direction. One or more battery packs can be arranged in each battery pack installation space.

[0064] Of course, in other embodiments, the battery pack mounting frame **4** may further include a middle partition frame for separating the battery packs **1** in the left-right direction. One or more

middle partition frames are provided to divide the interior of the battery pack mounting frame 4 into multiple battery pack installation spaces arranged in the left-right direction.

[0065] In one embodiment, referring to FIG. 2 and FIG. 3, the middle partition frame 4115 includes a top middle cross-beam 41151 (forming the cross-beam of the battery pack mounting frame 4), a left middle vertical beam 41152 (forming the vertical beam of the battery pack mounting frame 4), a right middle vertical beam 41153 (forming the vertical beam of the battery pack mounting frame 4) and a bottom supporting cross-beam 41154 (forming the cross-beam of the battery pack mounting frame 4). The top middle cross-beam 41151 and the bottom supporting cross-beam 41154 extend in the left-right direction of the vehicle body, and the left middle vertical beam 41152 and the right middle vertical beam 41153 extend in the up-down direction of the vehicle body; the top middle cross-beam 41151 is connected between the left upper longitudinal beam 419 and the right upper longitudinal beam 4111, and the top middle cross-beam 41151 is located between the front upper cross-beam 411 and the rear upper cross-beam 415 in the front-rear direction of the vehicle frame assembly; the bottom supporting cross-beam 41154 is connected between the left lower longitudinal beam 4110 and the right lower longitudinal beam 4112, and the bottom supporting cross-beam 41154 is located between the front lower cross-beam 415 and the rear lower cross-beam 616 in the front-rear direction of the vehicle frame assembly; the upper end of the left middle vertical beam 41152 is connected to the connection point of the top middle cross-beam 41151 and the left upper longitudinal beam 419, the lower end of the left middle vertical beam 41152 is connected to the connection point of the left lower longitudinal beam 4110 and the bottom supporting cross-beam 41154, the upper end of the right middle vertical beam 41153 is connected to the connection point of the top middle cross-beam 41151 and the right upper longitudinal beam 4111, and the lower end of the right middle vertical beam 41153 is connected to the connection point of the right lower longitudinal beam 4112 and the bottom supporting cross-beam 41154; the top middle cross-beam 41151, the left middle vertical beam 41152, the bottom supporting cross-beam 41154 and the right middle vertical beam 41153 are connected in sequence to form a middle closed-loop frame structure. The middle partition frame 4115 can enhance the middle strength of the battery pack mounting frame 4. Furthermore, the overall strength of the battery pack mounting frame 4 is further enhanced.

[0066] In one embodiment, referring to FIG. 2 and FIG. 3, the bottom supporting cross-beam 41154 includes a first supporting cross-beam 411541 and a second supporting cross-beam 411542. The first supporting cross-beam 411541 is located in front of the second supporting cross-beam 411542. The lower end of the left middle vertical beam 41152 is located between the left end of the first supporting cross-beam 411541 and the left end of the second supporting cross-beam 411542, and the lower end of the right middle vertical beam 41153 is located between the right end of the first supporting cross-beam 411541 and the right end of the second supporting cross-beam 411542. The bottom supporting cross-beam 41154 and the bottom middle longitudinal beam 4114 jointly support the bottom of the battery pack 1.

[0067] In one embodiment, referring to FIG. 2 and FIG. 3, the battery pack mounting frame 4 further includes a front supporting cross-beam 4116 (forming the cross-beam of the battery pack mounting frame 4) and a rear supporting cross-beam 4117 (forming the cross-beam of the battery pack mounting frame 4); the front supporting cross-beam 4116 is connected between the front end of the left lower longitudinal beam 4110 and the front end of the right lower longitudinal beam 4112, and the front supporting cross-beam 4116 is located behind the front lower cross-beam 412; the rear supporting cross-beam 4117 is connected between the rear end of the left lower longitudinal beam 4110 and the rear end of the right lower longitudinal beam 4112, and the rear supporting cross-beam 4117 is located in front of the rear lower cross-beam 416; the front supporting cross-beam 4116 and the first supporting cross-beam 411541 are used to support the bottom of a front-side battery pack 1, and the rear supporting cross-beam 4117 and the second supporting cross-beam 411542 are used to support the bottom of a rear-side battery pack 1. In this

way, the front and rear sides of the bottom of the front-side battery pack **1** are supported by the front supporting cross-beam **4116** and the first supporting cross-beam **411541**, and the bottom middle longitudinal beam **4114** further supports the bottom of the front-side battery pack **1**, so the front-side battery pack **1** is stably supported. Similarly, the front and rear sides of the bottom of the rear-side battery pack **1** are supported by the second supporting cross-beam **411542** and the rear supporting cross-beam **4117**, and the bottom middle longitudinal beam **4114** further supports the bottom of the rear-side battery pack **1**, so the rear-side battery pack **1** is stably supported.

[0068] If the battery pack mounting frame **4** is a multi-layer frame structure, the bottom of the upper-layer frame structure and the top of the lower-layer frame structure are combined into one to save materials and reduce the weight of the battery pack mounting frame **4**.

[0069] In one embodiment, referring to FIG. 2 and FIG. 3, the middle partition frame **4115** further includes a connecting vertical beam **41155** (forming the vertical beam of the battery pack mounting frame **4**). The connecting vertical beam **41155** extends in the up-down direction of the vehicle body. The upper end of the connecting vertical beam **41155** is connected to the junction of the top middle longitudinal beam **4113** and the top middle cross-beam **41151**, and the lower end of the connecting vertical beam **41155** is connected to the bottom middle longitudinal beam **4114** and is located between the first supporting cross-beam **411541** and the second supporting cross-beam **411542**. The connecting vertical beam **41155** can enhance the strength of the middle partition frame **4115**.

[0070] In one embodiment, referring to FIG. 2 and FIG. 3, there are two top middle longitudinal beams **4113**, two bottom middle longitudinal beams **4114** and two connecting vertical beams **41155**. The left top middle longitudinal beam **4113** is directly above the left bottom middle longitudinal beam **4114**, the right top middle longitudinal beam **4113** is directly above the right bottom middle longitudinal beam **4114**, the left connecting vertical beam **41155** is fixedly connected between the left top middle longitudinal beam **4113** and the left bottom middle longitudinal beam **4114**, and the right connecting vertical beam **41155** is fixedly connected between the right top middle longitudinal beam **4113** and the right bottom middle longitudinal beam **4114**. The top middle longitudinal beams **4113**, the bottom middle longitudinal beams **4114**, and the connecting vertical beams **41155** are each provided in pairs, further increasing the strength of the battery pack mounting frame **4**.

[0071] In one embodiment, the battery pack mounting frame **4** further includes multiple diagonal beams, and the diagonal beams are arranged between the intersection points of two longitudinal beams and cross-beams.

[0072] In a specific embodiment, referring to FIG. 2 and FIG. 3, the middle partition frame **4115** further includes a left diagonal beam **41156** (forming the diagonal beam of the battery pack mounting frame **4**) and a right diagonal beam **41157** (forming the diagonal beam of the battery pack mounting frame **4**); the upper end of the left diagonal beam **41156** is connected to the junction of the left top middle longitudinal beam **4113** and the top middle cross-beam **41151**, and the lower end of the left diagonal beam **41156** is connected to the left lower longitudinal beam **4110** and is located between the first supporting cross-beam **411541** and the second supporting cross-beam **411542**; the upper end of the right diagonal beam **41157** is connected to the junction of the right top middle longitudinal beam **4113** and the top middle cross-beam **41151**, and the lower end of the right diagonal beam **41157** is connected to the right lower longitudinal beam **4112** and is located between the first supporting cross-beam **411541** and the second supporting cross-beam **411542**. The left diagonal beam **41156** and the right diagonal beam **41157** can further enhance the strength of the middle partition frame **4115**.

[0073] In one embodiment, the battery pack mounting frame **4** further includes multiple mounting frame reinforcing plates, and the mounting frame reinforcing plates are arranged at the bottom and/or the side of the battery pack mounting frame.

[0074] In one embodiment, referring to FIG. 2 to FIG. 5, the multiple mounting frame reinforcing

plates include a left installation reinforcing plate **42** of the battery pack bracket and a right installation reinforcing plate **43** of the battery pack bracket. The left installation reinforcing plate **42** of the battery pack bracket and the right installation reinforcing plate **43** of the battery pack bracket extend in the front-rear direction of the vehicle body. The left installation reinforcing plate **42** of the battery pack bracket is detachably connected between the middle part of the left front vertical beam **413** and the middle part of the left rear vertical beam **417**, and the right installation reinforcing plate **413** of the battery pack bracket is detachably connected between the middle part of the right front vertical beam **414** and the middle part of the right rear vertical beam **418**. In this way, when the left installation reinforcing plate **42** of the battery pack bracket and the right installation reinforcing plate **43** of the battery pack bracket are provided, the battery pack **1** can be installed into the battery pack mounting frame **4** from the left side or the right side of the battery pack mounting frame **4**. The left installation reinforcing plate **42** of the battery pack bracket can strengthen the left side of the battery pack mounting frame **4** and, at the same time, can play a role in anti-collision on the left side of the battery pack **1**. The right installation reinforcing plate **42** of the battery pack bracket can strengthen the right side of the battery pack mounting frame **4** and, at the same time, can play a role in anti-collision on the right side of the battery pack **1**.

[0075] Of course, in other embodiments, one of the left installation reinforcing plate **42** of the battery pack bracket and the right installation reinforcing plate **43** of the battery pack bracket may be non-detachable.

[0076] In one embodiment, referring to FIG. 2 to FIG. 5, the multiple cross-beams of the battery pack mounting frame **4** further include a front middle cross-beam **4118** and a rear middle cross-beam **4119**. The front middle cross-beam **4118** and the rear middle cross-beam **4119** extend in the left-right direction of the vehicle body; the front middle cross-beam **4118** is connected between the middle part of the left front vertical beam **413** and the middle part of the right front vertical beam **414**, and the rear middle cross-beam **4119** is connected between the middle part of the left rear vertical beam **417** and the middle part of the right rear vertical beam **418**. The front middle cross-beam **4118** can strengthen the front side of the battery pack mounting frame **4**, and the rear middle cross-beam **4119** can strengthen the rear side of the battery pack mounting frame **4**.

[0077] In one embodiment, referring to FIG. 2 to FIG. 5, the battery pack mounting frame **4** further includes a left front connector **441**, a right front connector **442**, a left rear connector **443** and a right rear connector **444**. The left front connector **441** is fixed on the left front vertical beam **413** by bolts, riveting or welding, the right front connector **442** is fixed on the right front vertical beam **414** by bolts, riveting or welding, the left rear connector **443** is fixed on the left rear vertical beam **417** by bolts, riveting or welding, and the right rear connector **444** is fixed on the right rear vertical beam **418** by bolts, riveting or welding; the front end of the left installation reinforcing plate **42** of the battery pack bracket is fixed on the left front connector **441** by bolts, the rear end of the left installation reinforcing plate **42** of the battery pack bracket is fixed on the left rear connector **443** by bolts, the front end of the right installation reinforcing plate **43** of the battery pack bracket is fixed on the right front connector **442** by bolts, and the rear end of the right installation reinforcing plate **43** of the battery pack bracket is fixed on the right rear connector **444** by bolts.

[0078] Providing the left front connector **441**, the right front connector **442**, the left rear connector **443** and the right rear connector **444** can, on the one hand, increase the connection strength at the connection part, and on the other hand, facilitate the disassembly and assembly of the left installation reinforcing plate **42** of the battery pack bracket and the right installation reinforcing plate **43** of the battery pack bracket.

[0079] In addition, the second connector **52** can further be welded to the battery pack mounting frame **4**. In other words, the left front inner reinforcing plate **522**, the right front inner reinforcing plate **523**, the left rear inner reinforcing plate **524** and the right rear inner reinforcing plate **525** can further be welded to the battery pack mounting frame **4** to further increase the connection strength.

[0080] The left front inner reinforcing plate **522** and the right front inner reinforcing plate **523** can

be welded to the vehicle frame front section 2, and the left rear inner reinforcing plate 524 and the right rear inner reinforcing plate 525 can be welded to the vehicle frame rear section 3.

[0081] In one embodiment, referring to FIG. 9, the second connector 52 is further arranged with a downward-turned edge 526, and the downward-turned edge 526 extends inward along the left-right direction of the vehicle body. The downward-turned edge 526 of the left front inner reinforcing plate 522 supports the left side of the bottom of the vehicle frame front section 2 (the bottom surface of the left front longitudinal beam 21). The downward-turned edge 526 of the right front inner reinforcing plate 523 supports the right side of the bottom of the vehicle frame front section 4 (the bottom surface of the right front longitudinal beam 21); the downward-turned edge 526 of the left rear inner reinforcing plate 524 supports the left side of the bottom of the vehicle frame rear section 3 (the bottom surface of the left rear longitudinal beam 31); the downward-turned edge 526 of the right rear inner reinforcing plate 525 supports the right side of the bottom of the vehicle frame rear section 4 (the bottom surface of the right rear longitudinal beam 31). The downward-turned edge 526 can increase the contact area between the second connector 52 and the vehicle frame front section 2 and the vehicle frame rear section 3, and enhance the connection strength.

[0082] In one embodiment, the connection component 5 further includes a diagonal bracing beam, and the diagonal bracing beam connects the downward-turned edge 526 and the battery pack mounting frame 4. Specifically, the diagonal bracing beams of the multiple connection components 5 are divided into a left front diagonal bracing beam of the battery pack bracket, a right front diagonal bracing beam of the battery pack bracket, a left rear diagonal bracing beam of the battery pack bracket and a right rear diagonal bracing beam of the battery pack bracket; the upper end of the left front diagonal bracing beam of the battery pack bracket is fixed on the lower surface of the downward-turned edge 526 of the left front inner reinforcing plate 522 by bolts, riveting or welding, the lower end of the left front diagonal bracing beam of the battery pack bracket is fixed on the front side of the battery pack mounting frame 4 (the front lower cross-beam 412) by welding, and the left side of the left front diagonal bracing beam of the battery pack bracket is fixed on the right side of the left front outer reinforcing plate 532. The upper end of the right front diagonal bracing beam of the battery pack bracket is fixed on the lower surface of the downward-turned edge 526 of the right front inner reinforcing plate 523, the lower end of the right front diagonal bracing beam of the battery pack bracket is fixed on the front side of the battery pack mounting frame 4 (the front lower cross-beam 412), and the right side of the right front diagonal bracing beam of the battery pack bracket is fixed on the left side of the right front outer reinforcing plate 533; the left front diagonal bracing beam of the battery pack bracket and the right front diagonal bracing beam of the battery pack bracket can further increase the connection strength between the front side of the battery pack mounting frame 4 and the vehicle frame front section 2.

[0083] The upper end of the left rear diagonal bracing beam of the battery pack bracket is fixed on the lower surface of the downward-turned edge 526 of the left rear inner reinforcing plate 524, the lower end of the left rear diagonal bracing beam of the battery pack bracket is fixed on the rear side of the battery pack mounting frame 4 (the rear lower cross-beam 416), and the left side of the left rear diagonal bracing beam of the battery pack bracket is fixed on the right side of the left rear outer reinforcing plate 534; the upper end of the right rear diagonal bracing beam of the battery pack bracket is fixed on the lower surface of the downward-turned edge 526 of the right rear inner reinforcing plate 525, the lower end of the right rear diagonal bracing beam of the battery pack bracket is fixed on the rear side of the battery pack mounting frame 4 (the rear lower cross-beam 416), and the right side of the right rear diagonal bracing beam of the battery pack bracket is fixed on the left side of the right rear outer reinforcing plate 535. The left rear diagonal bracing beam of the battery pack bracket and the right rear diagonal bracing beam of the battery pack bracket can further increase the connection strength between the rear side of the battery pack mounting frame 4 and the vehicle frame rear section 3.

[0084] In one embodiment, referring to FIG. 6 and FIG. 7, the battery pack 1 is in the shape of a

“convex” character, in other words, notches **11** are arranged on both sides of the battery pack **1** along the vehicle body direction, and the notches **11** are used for the pipeline layout of the vehicle. A pipeline layout space is reserved at the notch **11** of the battery pack **1**, which is convenient for the pipeline layout of the whole vehicle. In addition, the notches **11** are located on the opposite sides of the battery pack **1**, so the pipelines are easier to disassemble, assemble and maintain, and it is convenient to arrange the connectors.

[0085] However, in other embodiments, the notch **11** can further be opened only on one side of the battery pack **1** in the left-right direction of the vehicle body. In other words, the notch **11** is opened only on one side of the battery pack **1** in the left-right direction of the vehicle body. At this time, it is equivalent to opening a notch **11** on one side of the square battery pack **1**.

[0086] In addition, referring to FIG. **10**, the energy storage structure provided by another embodiment of the present disclosure is different from the embodiment shown in FIG. **1** in that the battery pack **1** is square. At this time, in order to arrange the pipelines of the vehicle, the part of the top middle cross-beam **41151** located between the two top middle longitudinal beams **4113** is removed. The area between the two top middle longitudinal beams **4113** is used for the pipeline layout of the vehicle.

[0087] The first connectors **51** are fixedly connected to both the front and rear sides of the battery pack mounting frame **4**, and the first connectors **51** are arranged with avoidance holes **511** for the pipelines to pass through. Compared with the “convex”-shaped battery pack **1**, the square battery pack **1** does not need to open notches on the opposite sides, the internal space utilization rate of the square battery pack **1** is higher, more batteries can be accommodated inside, which is beneficial to increasing the capacity of the battery pack and improving the endurance of the whole vehicle.

[0088] In addition, an embodiment of the present disclosure provides a vehicle, which includes the energy storage structure of the foregoing embodiment.

[0089] The above are only the embodiments of the present disclosure and are not intended to limit the present disclosure. Any modifications, equivalent replacements, improvements, etc. made within the spirit and principles of the present disclosure shall be included in the protection scope of the present disclosure.

Claims

1. A vehicle frame assembly, comprising a vehicle frame front section, a battery pack mounting frame and a vehicle frame rear section which are sequentially connected along the length direction of the vehicle body, wherein the battery pack mounting frame is configured to install a battery pack.
2. The vehicle frame assembly according to claim 1, wherein along the width direction of the vehicle body, the size of the battery pack mounting frame is not less than the size of the vehicle frame front section or the vehicle frame rear section.
3. The vehicle frame assembly according to claim 2, wherein the vehicle frame assembly further comprises a connection component, the connection component connects the battery pack mounting frame with the vehicle frame front section or connects the battery pack mounting frame with the vehicle frame rear section.
4. The vehicle frame assembly according to claim 3, wherein the vehicle frame front section comprises a front cross-beam and two front longitudinal beams, the front cross-beam is connected between the two front longitudinal beams; the vehicle frame rear section comprises a rear cross-beam and two rear longitudinal beams, the rear cross-beam is connected between the two rear longitudinal beams; the connection component comprises a first connector and a second connector, the first connector connects the battery pack mounting frame with the front cross-beam or connects the battery pack mounting frame with the rear cross-beam, the second connector connects the battery pack mounting frame with the front longitudinal beam or connects the battery pack

mounting frame with the rear longitudinal beam.

5. The vehicle frame assembly according to claim 4, wherein the vehicle frame assembly further comprises a cover plate, the cover plate covers the top of the battery pack mounting frame, the second connector is arranged with an upward-turned edge for connecting with the cover plate.

6. The vehicle frame assembly according to claim 4, wherein the connection component further comprises a third connector, the third connector is arranged on the outer side of the second connector and connects the second connector and the battery pack mounting frame; the third connector is arranged with a side-turned edge, the side-turned edge is arranged outward along the width direction of the vehicle body to strengthen the strength of the third connector along the width direction of the vehicle body.

7. The vehicle frame assembly according to claim 1, wherein the battery pack mounting frame is formed by a plurality of longitudinal beams, a plurality of cross-beams and a plurality of vertical beams intersecting with each other to form a space for accommodating at least one layer of battery packs.

8. The vehicle frame assembly according to claim 1, wherein the battery pack mounting frame further comprises a plurality of diagonal beams and a plurality of mounting frame reinforcing plates, the diagonal beams are arranged between the intersection points of two longitudinal beams and cross-beams, the mounting frame reinforcing plates are arranged at the bottom or the side of the battery pack mounting frame.

9. An energy storage structure, comprising a battery pack and the vehicle frame assembly according to claim 1, wherein the battery pack is installed on the battery pack mounting frame.

10. The energy storage structure according to claim 9, wherein at least one of the two sides of the battery pack along the vehicle body direction is arranged with a notch, the notch is used for the pipeline layout on the vehicle.

11. An energy storage structure, comprising a battery pack and the vehicle frame assembly according to claim 2, wherein the battery pack is installed on the battery pack mounting frame.

12. The energy storage structure according to claim 11, wherein at least one of the two sides of the battery pack along the vehicle body direction is arranged with a notch, the notch is used for the pipeline layout on the vehicle.

13. An energy storage structure, comprising a battery pack and the vehicle frame assembly according to claim 3, wherein the battery pack is installed on the battery pack mounting frame.

14. The energy storage structure according to claim 13, wherein at least one of the two sides of the battery pack along the vehicle body direction is arranged with a notch, the notch is used for the pipeline layout on the vehicle.

15. An energy storage structure, comprising a battery pack and the vehicle frame assembly according to claim 4, wherein the battery pack is installed on the battery pack mounting frame.

16. The energy storage structure according to claim 15, wherein at least one of the two sides of the battery pack along the vehicle body direction is arranged with a notch, the notch is used for the pipeline layout on the vehicle.

17. An energy storage structure, comprising a battery pack and the vehicle frame assembly according to claim 8, wherein the battery pack is installed on the battery pack mounting frame.

18. The energy storage structure according to claim 17, wherein at least one of the two sides of the battery pack along the vehicle body direction is arranged with a notch, the notch is used for the pipeline layout on the vehicle.

19. A vehicle, comprising the energy storage structure according to claim 9.

20. A vehicle, comprising the energy storage structure according to claim 18.
